

# Data Requirements Document

## Online Shopping Platform (E-Commerce System)

### 1. Introduction

Once we listed out the main activities of the system, the next step was to figure out what data actually needs to be stored in the database to support those activities. We went through each activity and picked out the entities involved, along with the attributes we would need for each one. We have also marked the primary key for every entity, which is the field that uniquely identifies each record.

### 2. List of Entities

Based on the activities identified earlier (registration, browsing, ordering, payment and shipment tracking), the following entities were identified: Customer, Product, Category, Supplier, Order, Payment, Shipment, and Review. Each of these is described below with its attributes and primary key.

#### 2.1 Customer

| Attribute        | Type        | Notes                             |
|------------------|-------------|-----------------------------------|
| CustomerID       | Primary Key | Uniquely identifies each customer |
| Name             | Attribute   | Full name of the customer         |
| Email            | Attribute   | Used for login and notifications  |
| Phone            | Attribute   | Contact number                    |
| Password         | Attribute   | Login password (stored securely)  |
| Address          | Attribute   | Delivery address of the customer  |
| RegistrationDate | Attribute   | Date the customer signed up       |

#### 2.2 Product

| Attribute     | Type        | Notes                            |
|---------------|-------------|----------------------------------|
| ProductID     | Primary Key | Uniquely identifies each product |
| Name          | Attribute   | Name of the product              |
| Description   | Attribute   | Short description of the product |
| Price         | Attribute   | Selling price                    |
| StockQuantity | Attribute   | Number of units available        |
| CategoryID    | Foreign Key | Links to the Category entity     |
| SupplierID    | Foreign Key | Links to the Supplier entity     |

## 2.3 Category

| Attribute    | Type        | Notes                                  |
|--------------|-------------|----------------------------------------|
| CategoryID   | Primary Key | Uniquely identifies each category      |
| CategoryName | Attribute   | Name of the category, e.g. Electronics |
| Description  | Attribute   | Short description of the category      |

## 2.4 Supplier

| Attribute     | Type        | Notes                             |
|---------------|-------------|-----------------------------------|
| SupplierID    | Primary Key | Uniquely identifies each supplier |
| SupplierName  | Attribute   | Name of the supplier/company      |
| ContactNumber | Attribute   | Phone number of the supplier      |
| Email         | Attribute   | Email address of the supplier     |
| Address       | Attribute   | Supplier's address                |

## 2.5 Order

| Attribute   | Type        | Notes                                           |
|-------------|-------------|-------------------------------------------------|
| OrderID     | Primary Key | Uniquely identifies each order                  |
| CustomerID  | Foreign Key | Links to the Customer who placed the order      |
| OrderDate   | Attribute   | Date the order was placed                       |
| TotalAmount | Attribute   | Total value of the order                        |
| OrderStatus | Attribute   | Current status, e.g. placed, shipped, delivered |

## 2.6 Payment

| Attribute     | Type        | Notes                                      |
|---------------|-------------|--------------------------------------------|
| PaymentID     | Primary Key | Uniquely identifies each payment           |
| OrderID       | Foreign Key | Links to the Order being paid for          |
| PaymentMethod | Attribute   | Card, UPI, net banking or cash on delivery |
| PaymentStatus | Attribute   | Success, failed or pending                 |
| PaymentDate   | Attribute   | Date and time the payment was made         |
| AmountPaid    | Attribute   | Amount paid for the order                  |

## 2.7 Shipment

| Attribute      | Type        | Notes                                        |
|----------------|-------------|----------------------------------------------|
| ShipmentID     | Primary Key | Uniquely identifies each shipment            |
| OrderID        | Foreign Key | Links to the Order being shipped             |
| ShippingDate   | Attribute   | Date the order was shipped                   |
| DeliveryDate   | Attribute   | Expected or actual delivery date             |
| ShipmentStatus | Attribute   | Packed, shipped, out for delivery, delivered |
| TrackingNumber | Attribute   | Number used to track the shipment            |

## 2.8 Review

| Attribute  | Type        | Notes                                      |
|------------|-------------|--------------------------------------------|
| ReviewID   | Primary Key | Uniquely identifies each review            |
| ProductID  | Foreign Key | Links to the Product being reviewed        |
| CustomerID | Foreign Key | Links to the Customer who wrote the review |
| Rating     | Attribute   | Star rating given by the customer          |
| Comment    | Attribute   | Text of the review                         |
| ReviewDate | Attribute   | Date the review was posted                 |

## 3. Conclusion

These eight entities and their attributes should cover the data needed to support the main activities of the online shopping platform. The primary keys marked above will help uniquely identify each record, and the foreign keys show how the entities are related to one another, which will be useful when designing the actual database tables.